The Islamia University of Bahawalpur

Faculty of Computing

Department of (IT)



FINAL YEAR PROJECT PROPOSAL

Project Title: iClone — An Intelligent Virtual Human Assistant for Personalized Daily-Life Support

Submitted By : Atika Naheed
Registration Number(s): [Enter Registration Number(s)]
Submitted To : [Enter Supervisor Name]
Department: [Enter Department Name]
University: [Enter University Name]

1. Abstract

iClone is an intelligent virtual assistant designed to function as a personalized digital human replica that supports users in their daily activities. The system aims to provide a more natural and personalized interaction experience by combining artificial intelligence, natural language processing, voice interaction, personal knowledge management, and a visual digital avatar.

Unlike traditional virtual assistants that primarily respond to predefined commands, iClone will aim to understand user instructions, maintain relevant conversational context, learn user-approved preferences, and provide context-aware assistance. The system will support text-based and voice-based interaction, reminders, task management, personalized responses, and retrieval of information from user-provided documents and personal knowledge.

The proposed system will incorporate a digital avatar that represents the virtual assistant visually. Where technically feasible, the avatar will support speech synchronization to create a more human-like interaction experience. An optional voice-personalization feature may be implemented using the user's own voice samples with explicit consent.

The system will be developed as a web-based application using React.js for the frontend and Python with FastAPI for the backend. An appropriate large language model (LLM) will power conversational responses, while retrieval-augmented generation (RAG) will help the assistant retrieve relevant information from the user's approved knowledge base. A database will store user profiles, preferences, tasks, reminders, and conversation metadata.

The primary objective of iClone is to develop a practical, privacy-conscious AI assistant that offers personalized support through natural interaction while reducing the effort required to manage routine activities. The project will be evaluated through functional testing, usability assessment, response accuracy, and task-completion measurements.

Keywords: Artificial Intelligence, Virtual Assistant, Digital Human, Human Replica, Natural Language Processing, Voice Interaction, Large Language Models, Personalization, RAG.

2. Introduction

Artificial intelligence has significantly changed how people interact with computer systems. Traditional software applications generally require users to navigate menus, enter information manually, or issue specific commands to perform tasks. Modern AI-powered assistants provide an alternative by allowing users to communicate through natural language and receive relevant information or assistance.

However, many existing virtual assistants primarily focus on answering questions or executing individual commands. Their ability to maintain personalized context, interact through a human-like visual representation, and provide organized support across daily activities varies considerably.

The proposed iClone project aims to address this gap by developing a personalized virtual assistant that acts as a digital representation of an individual. The assistant will combine conversational AI, user-approved personal information, task management, voice interaction, and a visual avatar in a unified application.

The term *human replica* in this project refers to a digital representation that imitates selected aspects of a person's communication style, appearance, and voice, where authorized. It does not imply that the system possesses human consciousness, emotions, or independent human judgment.

The concept of a *silent assistant* means that iClone should assist without unnecessarily interrupting the user. It will provide discreet reminders, optional notifications, and context-aware suggestions while allowing the user to control when voice output, avatar interaction, and other notifications are activated.

The proposed system will initially focus on a manageable set of personal assistance features suitable for an undergraduate final year project.

3. Background and Motivation

People frequently manage multiple responsibilities, including appointments, daily tasks, personal notes, deadlines, and information retrieval. Remembering these responsibilities and repeatedly interacting with separate applications can become time-consuming.

Existing digital assistants can help with some of these activities, but a personalized digital human that combines conversational interaction, persistent user-approved context, task management, and visual representation provides an opportunity for further exploration.

Recent developments in large language models, speech recognition, speech synthesis, and retrieval-based information systems make it increasingly practical to develop such applications using existing technologies.

The motivation behind iClone is to investigate how these technologies can be integrated into a single assistant that is easy to use, personalized, and capable of supporting routine daily activities.

The project also provides an opportunity to study important software engineering and AI challenges, including response reliability, memory management, personalization, privacy, and usability.

4. Problem Statement

Individuals often depend on multiple digital applications to manage reminders, tasks, notes, personal information, and everyday queries. Conventional command-based systems may require users to learn specific commands or repeatedly provide information that has already been shared.

Furthermore, many assistant applications do not combine personalized conversational context, user-controlled personal knowledge, voice interaction, and a digital human representation within a single, consistent experience.

There is therefore an opportunity to develop an integrated AI-based virtual assistant that can understand natural-language requests, maintain relevant user-approved context, organize routine tasks, and provide personalized responses through text, voice, and a visual avatar.

The challenge is to implement these capabilities within a feasible undergraduate project while maintaining response reliability, user privacy, appropriate authorization, and manageable computational requirements.

iClone proposes to address this problem by integrating conversational AI, personal knowledge retrieval, task and reminder management, and a digital avatar into one application.

5. Aim of the Project

The main aim of this project is to design and develop iClone, an intelligent virtual human assistant that provides personalized daily-life assistance through natural-language interaction, user-approved memory, task management, voice capabilities, and a visual digital avatar.

6. Project Objectives

The specific objectives are:

1. To develop an interactive virtual assistant that understands and responds to natural-language text input.
2. To implement voice input and spoken responses using suitable speech recognition and text-to-speech technologies.
3. To create a customizable digital avatar representing the virtual assistant.
4. To develop a personalization mechanism that uses user-approved preferences and relevant conversational context.
5. To integrate a knowledge retrieval system that answers questions using user-uploaded documents and approved personal information.
6. To provide reminders, notes, task creation, task updates, and task-completion tracking.
7. To implement a discreet assistance mode that supports configurable notifications and minimizes unnecessary interruptions.

8. To establish secure user authentication and appropriate controls for personal data.
9. To evaluate the system in terms of functionality, response quality, usability, and task-completion performance.

8. Scope of the Project

8.1 Included Features

The initial version of iClone will include:

- User registration, login, and profile management.
- A conversational AI interface.
- Text-based communication with the assistant.
- Voice input and spoken AI responses.
- A customizable visual avatar.
- User-approved personalization and conversation context.
- Personal knowledge management using uploaded documents.
- Retrieval of relevant information from the knowledge base.
- Creation, modification, and deletion of tasks and reminders.
- Task completion tracking.
- Configurable notifications and discreet assistance settings.
- Conversation history and memory controls.
- Privacy settings and deletion of user data.
- Testing and evaluation of the implemented features.

8.2 Optional Advanced Features

The following features will be considered after the core system is functional:

- Voice cloning using voice samples supplied by the authorized user.
- Enhanced avatar animation and lip synchronization.
- More advanced personality customization.
- Additional language support.
- Context-aware suggestions based on explicitly enabled data sources.
- Integration with selected external calendar services.

Optional features will depend on development time, hardware capabilities, model compatibility, and licensing conditions.

8.3 Exclusions

The project will not attempt to create a conscious or emotionally sentient human replica. It will not guarantee perfect imitation of a real person's behavior or judgment.

The initial version will not include autonomous financial transactions, unrestricted computer control, medical diagnosis, or other high-risk actions. It will not continuously record users without their knowledge or permission.

The system will not require access to every application or sensor on the user's device. Any external integration will be limited to explicitly authorized functionality.

9. Proposed System

iClone will provide a unified interface through which users can communicate with a personalized virtual assistant.

A user will be able to enter a message or speak a request. The backend will process the input, retrieve relevant information from the user's authorized knowledge base, and generate an appropriate response.

If the request involves a supported action, such as creating a reminder or adding a task, the system will validate the request and execute the corresponding application function. The resulting information will be saved in the database.

The assistant will present its response through text and, when enabled, speech. A visual avatar may accompany the response with basic animation or lip synchronization.

The assistant will use only the personal information and integrations authorized by the user. Sensitive actions will require explicit confirmation where appropriate.

Example Scenario

A user opens iClone and says:

"Remind me to submit my assignment tomorrow at 10 AM."

The system will:

1. Convert the spoken request into text.
2. Identify the requested task and reminder time.
3. Ask for clarification if the date or time is ambiguous.
4. Save the task and reminder after validation.
5. Confirm the action through text or voice.
6. Display the task in the user's dashboard.
7. Notify the user at the configured time, subject to operating-system and application notification availability.

Another example is a user asking, "What tasks do I have pending today?" iClone will retrieve the user's relevant task records and provide a concise response.

10. Functional Requirements

The system shall provide the following functionality:

FR-01: Authentication
The system shall allow users to register, log in, and securely access their personal assistant.

FR-02: Conversational Interaction
The system shall accept text input and generate contextually relevant responses.

FR-03: Voice Input
The system shall convert supported spoken input into text.

FR-04: Speech Output
The system shall convert generated responses into speech when voice mode is enabled.

FR-05: Avatar Interaction
The system shall display a visual avatar and synchronize supported animations with the assistant's responses.

FR-06: Personalization
The system shall store and retrieve user-approved preferences and relevant personal context.

FR-07: Knowledge Retrieval
The system shall retrieve relevant content from uploaded documents and use it to answer user queries.

FR-08: Task Management
The system shall support task creation, editing, completion, and deletion.

FR-09: Reminders
The system shall allow users to schedule reminders and receive available notifications.

FR-10: Discreet Assistance
The system shall allow users to configure notification preferences and disable voice or visual interruptions.

FR-11: Privacy Controls
The system shall allow users to review, update, and delete supported personal information.

FR-12: Action Confirmation
The system shall validate supported actions and request confirmation before executing designated sensitive operations.

FR-13: Error Handling
The system shall handle invalid inputs, unavailable AI services, and failed operations with appropriate messages.

11. Non-Functional Requirements

Performance

The system should respond within a reasonable time under the available hardware and model configuration. Performance targets will be established during implementation and testing.

Usability

The interface should be simple, consistent, and accessible to users with different levels of technical experience.

Security

Passwords shall be stored using secure password hashing. Protected API endpoints shall enforce authentication and authorization.

Privacy

Personal information, voice recordings, uploaded documents, and conversation history shall be handled according to clearly defined permissions and retention settings.

Reliability

Task and reminder operations shall be validated before being saved. Failed operations shall not be reported as successful.

Maintainability

The application shall use modular components, documented APIs, and structured database access to facilitate testing and future development.

Portability

The initial system shall be designed for use in a modern web browser on a personal computer. Additional deployment targets may be considered later.

12. Proposed System Modules

Module 1: User Authentication and Profile Management

This module will handle registration, login, user preferences, and profile configuration. Each user will have access only to their authorized personal data.

Module 2: AI Conversation Engine

This module will receive user messages, maintain relevant conversational context, and generate responses through an appropriate language model.

The assistant's response style will be configurable. Personalization will rely on user-provided information and explicit preferences rather than an assumption that the system can automatically understand the user's entire personality.

Module 3: Personal Memory and Knowledge Base

This module will manage approved user preferences, relevant conversation summaries, and uploaded documents.

A retrieval-augmented generation pipeline will retrieve relevant passages from the knowledge base before generating answers. This can help ground responses in user-provided information, although it cannot eliminate all incorrect answers.

Users will be able to inspect and delete supported stored memories.

Module 4: Voice Interaction

This module will provide speech-to-text and text-to-speech functionality.

The first implementation will use a suitable existing speech recognition model and a compatible speech synthesis engine. Voice cloning may be included as an optional extension if its quality, licensing, and hardware requirements are manageable.

Module 5: Digital Human Avatar

This module will display the assistant as a visual digital character. The initial implementation may use a 2D avatar or lightweight 3D model.

Basic expressions, speaking animations, and lip synchronization will be added where technically feasible. A highly realistic, continuously animated human replica is not a mandatory requirement for the initial version.

Module 6: Daily Task and Reminder Manager

This module will allow users to create tasks, set deadlines, schedule reminders, and track completion status.

The system will provide a dashboard showing upcoming and pending tasks. Notification delivery will depend on the application's runtime environment and operating-system permissions.

Module 7: Silent Assistance and Notification Manager

This module will provide discreet notifications and configurable assistance preferences.

Users will be able to choose whether reminders appear visually, play a sound, or use spoken output. The assistant will not continuously listen to the user unless an explicitly enabled feature requires it.

Module 8: Security and Administration

This module will implement access control, secure handling of personal data, validation, error logging, and appropriate data deletion procedures.

An administrative interface, if required for the academic demonstration, will be limited to authorized system-management functions and will not expose private user conversations unnecessarily.

13. Proposed Technology Stack

Component	Proposed Technology
Frontend	React.js with TypeScript
UI Styling	Tailwind CSS
Backend	Python with FastAPI
AI Conversation	Suitable LLM through a local runtime or authorized API
Local AI Runtime	Ollama, where compatible with the selected model
Speech Recognition	Whisper or a suitable alternative
Speech Synthesis	Compatible local TTS engine
Avatar	Lightweight 2D/3D avatar with optional lip-sync
Database	SQLite for a single-user prototype or PostgreSQL for multi-user deployment
Knowledge Retrieval	Embeddings and a vector index
API Communication	REST API

Component	Proposed Technology
Authentication	Secure password hashing and token-based authentication
Testing	Pytest for backend testing and suitable frontend testing tools
Development Tools	Visual Studio Code or an AI-assisted development environment
Version Control	Git and GitHub

The final choice of models and libraries will depend on compatibility, licensing, performance, and hardware availability.

14. System Architecture

The proposed architecture consists of the following layers:

Layer 1: User Interface

The React frontend will provide the chat interface, avatar display, task dashboard, reminder controls, and user settings.

Layer 2: Backend API

FastAPI will handle authentication, request validation, task operations, memory management, and communication between the frontend and AI components.

Layer 3: AI Processing

The AI processing layer will include conversational response generation, contextual information retrieval, and supported speech-processing functions.

Layer 4: Application Services

Dedicated services will handle task creation, reminder scheduling, notifications, and user-approved actions.

Layer 5: Data Storage

The database will store user profiles, preferences, tasks, reminders, and conversation metadata. A document store or vector index will support knowledge retrieval where required.

Request Processing Workflow

1. The user submits a text or voice request.
2. The frontend sends the request to the backend.
3. The backend authenticates the user and validates the input.
4. Relevant conversation context or knowledge-base information is retrieved.

5. The AI engine generates a response or identifies a supported action.
6. The backend validates and executes the action when authorized.
7. Results are saved where appropriate.
8. The frontend displays the response and optionally plays speech or animates the avatar.

For tasks involving irreversible or sensitive operations, the system will require appropriate confirmation before execution.

15. Preliminary Database Design

The following tables are proposed for the initial implementation.

15.1 Users

Stores account information.

- user_id — Primary Key
- name
- email
- password_hash
- created_at

15.2 User_Preferences

Stores personalization settings.

- preference_id — Primary Key
- user_id — Foreign Key
- preference_key
- preference_value
- updated_at

15.3 Conversations

Stores conversation session metadata.

- conversation_id — Primary Key
- user_id — Foreign Key
- title
- created_at

15.4 Messages

Stores conversation messages according to the user's retention settings.

- message_id — Primary Key
- conversation_id — Foreign Key
- sender_type
- message_content
- created_at

15.5 Memories

Stores user-approved persistent information.

- memory_id — Primary Key
- user_id — Foreign Key
- memory_content
- memory_category
- created_at
- updated_at

15.6 Documents

Stores metadata about uploaded knowledge documents.

- document_id — Primary Key
- user_id — Foreign Key
- file_name
- file_path or storage_reference
- processing_status
- uploaded_at

15.7 Tasks

Stores daily tasks.

- task_id — Primary Key
- user_id — Foreign Key
- title
- description
- due_date
- status
- created_at

15.8 Reminders

Stores scheduled reminders.

- reminder_id — Primary Key
- user_id — Foreign Key

- task_id — Optional Foreign Key
- reminder_time
- notification_type
- status
- created_at

15.9 Avatar_Settings

Stores avatar configuration.

- avatar_id — Primary Key
- user_id — Foreign Key
- avatar_type
- avatar_asset_reference
- voice_mode
- created_at

15.10 Notification_Preferences

Stores notification and discreet-assistance settings.

- notification_preference_id — Primary Key
- user_id — Foreign Key
- sound_enabled
- voice_enabled
- visual_notifications_enabled
- quiet_hours_start
- quiet_hours_end

This is a preliminary schema. Indexes, database constraints, document-chunk metadata, and additional tables will be defined during implementation. Voice recordings and other sensitive media will not be retained unnecessarily.

16. Development Methodology

The project will follow an iterative Agile development approach. Development will be divided into small increments so that each major feature can be implemented and tested before the next is added.

Phase 1: Requirement Analysis

The project requirements, user scenarios, functional requirements, and technical constraints will be identified. The initial scope will prioritize conversational AI, personal memory, task management, and basic avatar interaction.

Phase 2: System Design

The team will prepare the system architecture, database schema, interface wireframes, API definitions, and use-case diagrams.

Phase 3: Core Application Development

The frontend, backend, authentication system, database, and conversational interface will be implemented.

Phase 4: AI and Personalization Integration

The selected language model will be integrated. User-approved memory and document-based knowledge retrieval will be implemented and tested.

Phase 5: Voice and Avatar Integration

Speech recognition, speech synthesis, and a basic avatar will be integrated. Advanced voice cloning or lip synchronization will be attempted only after the core system is stable.

Phase 6: Task Management and Notifications

Task management, reminder scheduling, notification preferences, and discreet assistance behavior will be implemented.

Phase 7: Testing and Evaluation

The system will undergo functional, integration, usability, and performance testing. Errors and limitations will be documented and addressed.

Phase 8: Documentation and Final Demonstration

The final report, user manual, diagrams, test results, presentation, and project demonstration will be prepared.

17. Project Timeline

The following is a proposed 16-week schedule.

Activity	Duration
Requirement gathering and problem analysis	Weeks 1–2

Activity	Duration
Literature review and feasibility analysis	Weeks 2–3
System architecture and database design	Weeks 3–4
UI/UX design and authentication	Weeks 4–5
Backend and conversational interface	Weeks 5–7
AI integration and personal memory	Weeks 7–9
Knowledge-base and document retrieval	Weeks 9–10
Task management and reminders	Weeks 10–11
Voice interaction	Weeks 11–12
Avatar integration	Weeks 12–13
Security, integration, and testing	Weeks 13–14
Evaluation and improvements	Weeks 14–15
Final documentation and presentation	Week 16

Some activities will overlap. If development time becomes limited, the priority will be to complete the core assistant, memory, task management, and basic voice interaction before advanced avatar features.

18. Testing and Evaluation Plan

The project will be evaluated using predefined test cases and measurable criteria.

18.1 Functional Testing

Tests will verify that:

- Users can register and log in.
- Text requests receive appropriate responses.
- Approved preferences can be saved and retrieved.
- Uploaded documents can be processed and searched.
- Tasks can be created, updated, completed, and deleted.
- Reminders are stored correctly and triggered under supported runtime conditions.
- Voice input and output function with supported devices.
- Privacy settings are enforced.

18.2 AI Response Evaluation

A test dataset will be prepared containing questions about uploaded documents, saved preferences, task information, and general assistant interactions.

Responses will be assessed for relevance, factual correctness, and whether they are grounded in the available information.

18.3 Performance Evaluation

The team will measure response latency, task-operation success rates, and resource consumption under the available hardware configuration.

18.4 Usability Evaluation

A small group of volunteer users will perform representative tasks, such as creating reminders, retrieving personal information, and asking questions through voice or text.

Feedback will be collected using a short questionnaire and observations of task completion.

18.5 Proposed Success Criteria

The following are initial targets, not already achieved results:

- At least 95% successful execution of predefined core functional test cases.
- At least 85% correct responses on a curated set of answerable questions grounded in the test knowledge base.
- At least 90% successful completion of supported task-management test scenarios.
- Successful end-to-end demonstration of text interaction, personal context retrieval, and task reminders.
- No unresolved critical authentication or authorization defects at final submission.

The results will be reported transparently, including failures, limitations, and the size of the evaluation dataset.

19. Feasibility Analysis

19.1 Technical Feasibility

The project is technically feasible using existing web-development frameworks, language models, speech-processing tools, and avatar technologies.

However, realistic voice cloning and high-quality real-time digital human animation introduce additional complexity. A basic avatar and standard speech synthesis are therefore recommended for the initial implementation.

19.2 Economic Feasibility

The project can be developed primarily with open-source software and local AI models, reducing dependence on paid API services.

Potential costs include computer hardware, electricity, optional cloud hosting, storage, and services required for advanced voice or avatar features. The actual budget will depend on the chosen tools and available equipment.

19.3 Operational Feasibility

The application will provide a web interface familiar to most users. A dashboard, conversational interface, and straightforward task controls should make the system relatively easy to operate.

19.4 Schedule Feasibility

The core system is achievable within a typical undergraduate project schedule if the scope remains controlled. Advanced digital-human realism should be treated as an optional extension rather than a dependency for completing the project.

20. Privacy, Security, and Ethical Considerations

Because iClone may process personal information, voice samples, uploaded documents, and conversation history, privacy and responsible use are important design requirements.

The project will follow these principles:

1. **Informed consent:** The system will obtain permission before processing voice samples or personal media for replica creation.
2. **Authorized identity:** Voice cloning and likeness replication will be restricted to the user or another person who has provided appropriate authorization.
3. **Data minimization:** Only information needed for the supported features will be stored.
4. **User control:** Users will be able to review and delete supported memories and personal records.
5. **Secure authentication:** Passwords will be securely hashed, and private API routes will require authorization.
6. **Transparent identity:** The application will clearly identify the assistant as AI-generated rather than representing it as the actual person.
7. **Controlled actions:** Sensitive actions will require appropriate confirmation.
8. **No unauthorized surveillance:** Continuous recording or background monitoring will not be enabled without explicit user control.
9. **Responsible model use:** Third-party models and libraries will be used according to their applicable licenses and restrictions.
10. **Secure storage:** Sensitive information and credentials will not be exposed in client-side code or public repositories.

21. Expected Outcomes

At the end of the project, the following outcomes are expected:

- A working web-based virtual assistant application.
- A conversational interface powered by an AI language model.
- Personalized responses based on user-approved information.
- A memory and knowledge-retrieval mechanism.
- A task and reminder management system.
- Voice-based interaction for supported input and output.
- A basic visual digital human avatar.
- Configurable notifications and discreet-assistance settings.
- Authentication, privacy controls, and structured data storage.
- A documented testing and evaluation report.
- Complete project documentation and a final demonstration.

The final system will demonstrate how multiple AI capabilities can be integrated into a practical personal-assistance application.

22. Limitations

The system may produce incorrect or incomplete responses because language models do not guarantee factual accuracy.

Personalization will depend on the information users provide and the quality of the memory and retrieval mechanisms.

Speech recognition may perform differently across accents, background-noise conditions, and languages.

Realistic voice cloning and avatar animation may require substantial computational resources and may not achieve professional commercial quality.

Notifications may depend on browser permissions, operating-system settings, and whether the application is running.

The system will not be considered a replacement for human judgment in sensitive or high-risk decisions.

23. Future Enhancements

Future versions of iClone may include:

- More realistic 3D human avatars.
- Improved real-time lip synchronization and facial expressions.
- Authorized personalized voice cloning.
- Additional language support.
- Integration with calendar, email, and productivity services.
- Optional desktop and mobile applications.
- User-approved context from additional data sources.
- Improved long-term memory management.
- More sophisticated scheduling and proactive assistance.
- Multi-device synchronization and secure cloud backup.
- Further evaluation of personalization quality and long-term usability.

These enhancements will be considered after the core objectives have been achieved and validated.

24. Conclusion

iClone proposes an intelligent virtual human assistant that combines conversational AI, personalization, voice interaction, task management, and a visual avatar to support users in everyday life.

The project addresses the need for a unified and personalized assistant that can retrieve relevant information, maintain user-approved context, organize routine tasks, and communicate naturally without unnecessary interruptions.

By using established software frameworks and suitable AI technologies, the project aims to deliver a functional and demonstrable undergraduate system while investigating important issues related to response reliability, usability, privacy, and human-like interaction.

The project will prioritize a reliable core implementation before attempting advanced features such as realistic voice cloning and highly detailed digital-human animation. Its success will be evaluated through documented tests and measurable results rather than visual appearance alone.